

CampusCircle

A Prototype-Based Research Study on a
Campus Marketplace and Resource
Sharing Platform.

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ABSTRACT

This paper presents CampusCircle, a full-stack web application designed to address a recurring and measurable problem in engineering colleges across India: first-year students spend between ₹3,000 and ₹6,000 in the first two weeks of their academic life on materials, drafting instruments, textbooks, lab coats, and scientific calculators, that their seniors have already purchased and no longer use. The platform operationalises a borrow-first philosophy, connecting verified students within the same institution through a structured, trust-rated lending and knowledge-sharing system. This paper documents the problem identification process, original survey-based research and validation, product design decisions, technical architecture, challenges encountered, and projected impact. It is written from the perspective of the founding student developer and is intended both as an academic submission and as a publicly shareable founder's document.

Keywords: peer-to-peer lending, campus marketplace, student sustainability, full-stack development, borrow economy, trust systems, engineering education

1. Introduction

The first two weeks of an engineering student's college life are a period of acute financial pressure and informational chaos. Without guidance, students often buy everything they can find, driven by anxiety rather than actual need. What makes this pattern particularly wasteful is that every item a first-year student purchases in panic has already been purchased by a senior, and is likely sitting unused in a bag, cupboard, or corner of a hostel room.

This paper documents the research, design, and development of CampusCircle: a platform built specifically to solve this problem at Anjuman-I-Islam Kalsekar Technical Campus (AIKTC), Navi Mumbai. The solution is not a marketplace for second-hand goods. It is a structured borrowing network, a system where the default transaction is: 'I have this. You can use it. Return it when you are done.'

The core insight driving this project is simple: the problem is not a shortage of resources. The problem is a broken connection between people who have things and people who need them.

1.1 Motivation

As a student at AIKTC, the researcher observed this pattern directly: seniors with unopened drafter kits, juniors spending ₹850 on a rolling scale they would use for one subject over four months. The observation prompted a formal investigation: was this an isolated observation, or a systemic problem? The survey findings (Section 3) confirmed it was systemic.

The secondary motivation was sustainability. Engineering education generates a significant volume of single-use academic purchases, items used for one semester, one subject, or fewer than ten sessions. A borrowing platform directly reduces this cycle of consumption without requiring any sacrifice from participants.

1.2 Scope

This paper covers:

- (1) the problem and its dimensions
- (2) primary research conducted through student surveys
- (3) the product decisions made in response to research findings
- (4) the technical architecture of the platform
- (5) challenges and critical objections
- (6) impact measurement and future scope.

2. Problem Statement

Every academic year, first-year engineering students at institutions like AIKTC walk into college without a clear understanding of what they actually need. In the absence of guidance, they purchase items based on general advice, peer anxiety, or aggressive marketing by stationery vendors positioned at college gates.

2.1 The Expenditure Pattern

A typical first-year engineering student in Maharashtra purchases the following items in their first two weeks:

Items	Market Price (₹)	Sessions Used	Cost Per Use (₹)
Full drafter kit (compass, set squares, divider)	₹1,200 – ₹1,800	8–12	₹120 – ₹225
White lab coat	₹350 – ₹500	10–12 labs	₹35 – ₹50
Casio fx-991ES Plus scientific calculator	₹1,100 – ₹1,400	Reusable	—
Graph paper pads, drawing sheets, stationery, Assignment sheets, Unused Files	₹900 – ₹1,500	Partial Use	—
Total Estimated Expenditure	₹3,500 – ₹5,200	—	—

Table 1: Typical first-week purchases by a first-year engineering student. Prices sourced from local vendors and online platforms, cross-verified with survey respondents.

2.2 The Structural Gap

The problem is structural, not individual. Seniors students in Semester 3, 5, or 7 own all of the above items. Many of these items are in good or like-new condition. Yet no reliable channel exists to connect these students with juniors who need them.

Existing informal mechanisms fail in predictable ways:

- WhatsApp groups: unverifiable, chaotic, and require social capital the first-year student does not yet have
- Word of mouth: entirely luck-based and reaches a tiny fraction of potential connections
- College notice boards: static, one-directional, not searchable or filterable
- Facebook Marketplace / OLX: not campus-specific; introduces trust and logistics problems

The result is that two realities coexist on the same campus simultaneously: seniors with idle resources, and juniors spending money to acquire the same resources, new.

2.3 The Trust Problem

A secondary but equally important dimension of the problem is trust. Even when informal connections exist, a junior finding out through a friend that a senior has a spare lab coat, the transaction carries uncertainty. There is no mechanism to verify the item's condition, to hold either party accountable, or to build the kind of reputation that makes repeated transactions safe. Campus Circle's trust and rating system is a direct design response to this gap.

3. Research and Validation

Before building, a structured validation process was conducted to confirm that the problem was real, widespread, and not already solved by an existing mechanism.

3.1 Survey Methodology

A Google Form survey was distributed to students across Semester 1 through Semester 7 at AIKTC. The survey was designed to gather quantitative data on spending patterns, awareness of existing exchange mechanisms, and willingness to participate in a structured borrowing platform.

Key questions included:

- Approximate total spend on academic materials in the first two weeks of college
- Which items were purchased and whether they are still regularly used
- Awareness of any campus-specific mechanism for borrowing or exchanging academic items
- Whether the respondent had idle items they would be willing to lend
- Primary concerns about borrowing from or lending to a fellow student

3.2 Key Survey Findings

The survey responses validated the initial hypothesis across several dimensions:

Findings	Response
Students who spent ₹3,000+ in first 2 weeks	~74% of Sem 1 respondents
Students with unused drafter kits (Sem 3+)	~81% of senior respondents
Awareness of any campus borrowing channel	~12% (almost none)
Seniors willing to lend if a trusted platform existed	~89% said yes or maybe
Primary concern about lending: item not returned	~67% cited this
Primary concern about borrowing: item condition	~58% cited this

Table 2: Summary of key survey findings. Percentages are approximate and based on the available sample of respondents at AIKTC.

3.3 Critical Objections Received

The survey and informal feedback sessions also surfaced several critical objections to the proposed solution. These are documented here because they directly shaped product decisions:

Objection 1: "Why not just buy new? It is not that expensive."

Response: ₹3,000–₹6,000 is a significant sum for many students, particularly those from non-metropolitan backgrounds. More importantly, the question inverts the problem: the items already exist and are idle. The environmental and financial cost of manufacturing new items when equivalent ones are sitting unused is the precise inefficiency that CampusCircle addresses.

Objection 2: "Nobody wants to buy used stuff."

Response: This concern is based on the assumption that students always prefer new products. However, many college-use items such as lab equipment, drawing boards, calculators, and reference books are used only for one or two semesters. Students often look for affordable alternatives instead of purchasing expensive new materials for short-term academic use. CampusCircle enables seniors to pass down usable resources at lower costs, making education more accessible and reducing unnecessary waste.

Objection 3: "Nobody would buy notes or assignment papers."

Response: CampusCircle is not designed to commercialize academic notes or assignments. Instead, the platform encourages graduating seniors to voluntarily share useful materials such as past-year question papers, assignment references, and practical guidance for juniors. These resources are intended to support learning, revision, and academic preparation rather than direct selling. The goal is to preserve valuable student-generated knowledge that would otherwise be discarded after graduation.

Objection 4: "What if someone does not return the item?"

Response: This is the most legitimate concern and the one that generated the most design effort. CampusCircle's trust system, return reminders, overdue escalation to admin, and temporary account suspension for non-return are all direct responses to this objection. The platform does not assume goodwill, it structures and incentivises it.

4. Solution Design - CampusCircle

4.1 Core Philosophy

CampusCircle is explicitly not a buy-sell marketplace. The foundational design decision made after reviewing the survey data and consulting with potential users was to make borrowing the default and selling an opt-in exception. This distinction matters because it changes the social contract of the platform: participants are contributing to a shared ecosystem, not conducting isolated commercial transactions.

The platform's philosophy can be stated as: *Every item has already served its owner. The most valuable thing that the owner can now do is let it serve someone else.*

4.2 Key Features

The platform is structured around eleven features, each tied directly to a validated user need:

Feature	Description	Problem Addressed
College-Verified Onboarding	Registration gated by institutional email. Branch and semester declared at setup.	Trust & identity verification
Personalized Borrow Feed	Home feed filtered by the logged-in student's branch and semester automatically.	Relevance — ECS Sem 1 sees ECS Sem 1 items
Borrow Request Flow	Junior requests with context; senior accepts, declines, or counters. Status tracked end-to-end.	Structured, accountable transaction
Notes & Study Material Library	Seniors upload branch/semester-tagged notes. Always free. Knowledge Contributor badges awarded.	Crowdsourced senior knowledge
Trust & Reputation System	Both parties rated after each completed borrow. Trust Score displayed publicly.	Return accountability

Return Reminder & Overdue Handling	Daily cron job; reminders 3 days before due date; admin escalation at 5 days overdue.	Non-return risk mitigation
Starter Kit Guide	Semester and branch-specific guide by seniors: what to borrow, buy, or skip.	Panic buying prevention
ML Recommendation Engine	FastAPI microservice using TF-IDF + cosine similarity to personalise the home feed.	Discovery of relevant items

Table 3: CampusCircle features mapped to the validated problems they address.

5. Technical Architecture

5.1 Technology Stack Overview

Layer	Language	Framework	Key Libraries
Frontend UI	JavaScript (JSX)	React 18 + Vite	TailwindCSS, Framer Motion
Frontend Routing	JavaScript	React Router v6	—
Backend API	JavaScript (Node)	Express.js	JWT, bcryptjs, Nodemailer
Real-Time Layer	JavaScript	Socket.io	node-cron
Database	MQL (JSON-like)	MongoDB	Mongoose ODM
ML Microservice	Python	FastAPI	scikit-learn, pandas
File Storage	—	Cloudinary CDN	Cloudinary SDK
Containerisation	YAML / Shell	Docker Compose	Docker

Table 4: Full technology stack used in CampusCircle.

5.2 System Architecture

CampusCircle follows a microservice-adjacent architecture with clear separation of concerns across four services:

- Client service (React + Vite): all user interface and client-side state
- Server service (Node.js + Express): REST API, authentication, real-time WebSocket, and scheduled jobs
- ML service (Python + FastAPI): recommendation endpoint, called internally by the server — never directly by the client
- Storage service (Cloudinary): all binary file storage; the backend generates signed upload URLs, actual bytes travel directly from browser to CDN

This separation means the frontend is never burdened with ML logic, and the ML service can be updated independently without affecting the rest of the stack.

5.3 Recommendation Engine

The ML recommendation engine is a FastAPI microservice exposing a single / suggested endpoint. It accepts a student's branch and semester and returns a ranked list of item IDs based on historical borrow patterns.

The algorithm begins with content-based filtering using TF-IDF vectorisation and cosine similarity across item descriptions and category metadata. As the platform accumulates more borrowed data, the algorithm is designed to evolve toward collaborative filtering, finding students with similar borrow histories and surfacing items borrowed by others like them.

This progression from content-based to collaborative filtering mirrors the platform's growth: in its early stage (limited data), it uses item characteristics; as data accumulates, it uses human behaviour patterns.

5.4 Trust System Implementation

The trust and reputation system is implemented across four platform layers: the rating widget (frontend), the rating API endpoint (backend), the trust score calculator (MongoDB aggregation), and the cron-based overdue handler (scheduled job).

After every completed borrow cycle, both parties are prompted to rate each other on three dimensions each. Seniors are rated on item accuracy, responsiveness, and helpfulness. Juniors are rated on return timeliness and care of the item. Composite trust scores are publicly visible on every profile, creating a transparent reputation economy that incentivises good behaviour without requiring enforcement in every individual transaction.

6. Challenges and Lessons Learned

6.1 Technical Challenges

Real-Time Chat Architecture: Implementing Socket.io for peer-to-peer chat within a borrow thread required careful session management. The challenge was ensuring that when a new user connected, they received the full unread message history, not just future messages. This was solved by persisting all messages in MongoDB and loading thread history on socket join.

ML Cold Start Problem: The recommendation engine faces the classic cold start problem: with few users and few borrows, collaborative filtering cannot generate meaningful recommendations. The solution is a two-phase algorithm, content-based filtering in the early stage, transitioning to collaborative filtering as data accumulates. The content-based phase uses item metadata (branch, semester, category) to generate relevant suggestions even with zero borrow history.

Trust Without Enforcement: Designing a trust system that works without legal enforcement is a genuine challenge. The solution is layered: soft reminders via cron job, escalation to admin, and temporary borrowing suspension for overdue accounts. No single mechanism is sufficient; all three working together create a meaningful disincentive for non-return.

6.2 Human and Behavioural Challenges

Senior Participation: The platform's supply side depends entirely on seniors voluntarily listing items. Early user research indicated that seniors were willing to lend the friction was the effort of listing. The product response was to make listing as fast as possible: name, category, condition, up to 3 photos, and a duration. No price field by default. The opt-in selling toggle was added specifically to not complicate the listing flow for the majority of lenders who just want to lend.

Philosophical Resistance: Some users questioned whether a borrow-first model could sustain senior motivation. 'What do I get from lending?' is a legitimate question. The answer the platform offers is: trust score, Knowledge Contributor badges, and the satisfaction of seeing 'Has helped 14 juniors this semester' on a profile. Whether intrinsic motivation is sufficient at scale is an open question the platform has not yet answered definitively.

6.3 What Would Be Done Differently

- Start the survey earlier and with a larger sample the current data is directionally correct but would be stronger with higher respondent numbers
- Build the Starter Kit Guide feature first, not last it is the highest-value, lowest-friction feature for new users and should have anchored the initial onboarding experience
- Design the admin dashboard for a real administrator from day one the current dashboard was designed around demo metrics rather than actual administrative workflows

7. Future Scope & Conclusion

7.1 Future Scope

- Expand to all branches at AIKTC, with branch-specific starter kit guides contributed by seniors across departments
- Introduce a gifting layer — items that seniors want to donate permanently rather than lend
- Pilot the model at a second engineering college to test whether the community dynamics are replicable in a different institutional context
- Build an iOS and Android app to reduce friction for mobile-first users
- Introduce sustainability reporting: total items kept out of new purchase cycles, estimated CO2 equivalent saved from reduced manufacturing
- Develop collaborative filtering fully as borrow data accumulates, moving beyond content-based recommendations

7.2 Conclusion

CampusCircle was built to solve a problem that is simultaneously invisible and universal. Every engineering college in India has seniors with idle drafters and juniors spending money to buy new ones. The gap is not a failure of goodwill, it is a failure of connection. The platform demonstrates that a structured, trust-rated, college-verified borrowing network can close this gap and generate measurable financial and social value for its users.

The deeper lesson from this project is about what a founder's research actually looks like at the student level. It does not require venture backing or a large team. It requires observing a real problem closely enough to understand its structure, talking to the people who experience it, taking their objections seriously enough to let those objections reshape the product, and then building something that makes the connection happen.

At the same time, this project also highlighted the importance of teamwork in problem-solving. Despite different ideas, challenges in coordination, and varying perspectives within the team, the development of CampusCircle became possible through collaborative effort, shared discussions, feedback collection, and continuous refinement of the concept. Every obstacle, criticism, and suggestion contributed to shaping the platform into a more practical and student-focused solution. CampusCircle is not just a marketplace idea; it is the result of collective learning, adaptation, and the attempt to solve a genuine campus-level problem through innovation and collaboration.

8. References

The following resources informed the design, research methodology, and technical implementation of CampusCircle:

1. Anjuman-I-Islam Kalsekar Technical Campus - Academic Calendar and Course Structure, CSE Department, 2025–26
2. CampusShare Full Stack Application Documentation - Internal project document, authored by the founding developer, 2026
3. Google Forms - Survey instrument used for student needs assessment and validation (administered at AIKTC, 2026)
4. Socket.io Documentation - Real-time bidirectional event-based communication.
<https://socket.io/docs/>
5. FastAPI Documentation - Building high-performance APIs with Python.
<https://fastapi.tiangolo.com/>